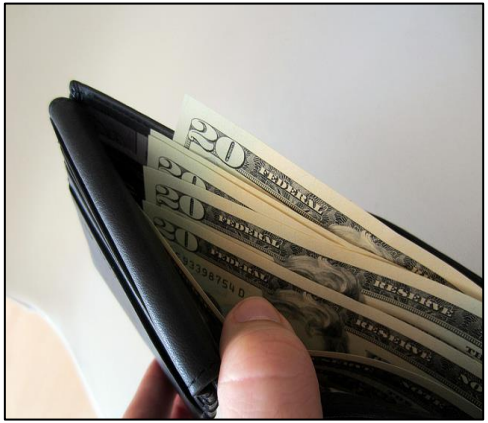
Research Topics in Blockchain
&
Blockchain and Its Application



How to Store and use Bitcoins Wallets

Wallets

- ✓ If you're storing your bitcoins locally, you'd typically use wallet software, which is software that keeps track of all your coins, manages all the details of your keys, and makes things convenient with a nice user interface.
- ✓ If you want to send $4.25 worth of bitcoins to your local coffee shop the wallet software would give you some easy way to do that. Wallet software is especially useful because you typically want to use a whole bunch of different addresses with different keys associated with them.
- ✓ As you may remember, creating a new public/private key pair is easy, and you can utilize this to improve your anonymity or privacy. Wallet software gives you a simple interface that tells you how much is in your wallet. When you want to spend bitcoins, it handles the details of which keys to use and how to generate new addresses and so on.



In the context of **cryptocurrencies**, encoding is used to represent **public keys, private keys, and addresses** in a human-readable format.

Encoding keys: base 58 and QR codes .To spend or receive bitcoins, you also need a way to exchange an address with the other party the address to which bitcoins are to be sent. There are two main ways in which addresses are encoded so that they can be communicated from receiver to spender: as a text string or as a QR code.

1. What is Base58 Encoding?

- **Base58** is a modified version of Base64, used in Bitcoin and other cryptocurrencies to encode addresses and private/public keys.
- It **removes easily confused characters** (like 0, O, I, and l) to reduce errors in manual entry.
- Bitcoin addresses (and MultiChain addresses) are typically encoded using Base58.

Example of a Base58 Bitcoin Address:

```
1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa
```

2. QR Codes for Crypto Addresses

- Instead of manually typing long Base58 addresses, they can be converted into **QR codes**.
- A **QR code** is a machine-readable representation of the address, making it easier to scan and share.
- This is commonly used in **wallet apps** for quick payments.

Example of Usage:

- When sending or receiving Bitcoin, you **scan a QR code** instead of typing the long address.

`1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa`

The address that received the very first Bitcoin block reward in the genesis block, base58 encoded.



Figure 4.1: **a QR code representing an actual Bitcoin address.** Feel free to send us some bitcoins.

What is a Vanity Address?

A **vanity address** is a specially crafted cryptocurrency or blockchain address that contains a **specific sequence of characters** (such as a name or phrase) chosen by the user. It is created by **generating many addresses** and selecting the one that matches the desired pattern.

For example, a Bitcoin or MultiChain address typically looks like this:

`1J7bGvM9KqF9K5X2hS7P3u2a6Hk6Y3iFZ`

A **vanity address** could look like:

`1YourNameXyz1234abcd...`

`1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa`

The address that received the very first Bitcoin block reward in the genesis block, base58 encoded.



Figure 4.1: **a QR code representing an actual Bitcoin address.** Feel free to send us some bitcoins.

How to Store and Use ~~Bitcoins~~ Secret Keys

Bitcoin wallets require several key components to ensure **security, functionality, and usability**. Below are the fundamental elements:

1.Address Public and Private Keys

Pair of cryptographic keys:

- ✓ **Public Key** → Shared with others to receive Bitcoin.
- ✓ **Private Key** → Kept secret to sign transactions.

Hierarchical deterministic (HD) wallets (BIP32/BIP44)

Hardware & Cold Storage Support

Hardware wallets store private keys **offline** to prevent hacking.

Cold wallets are used for long-term storage, while **hot wallets** handle frequent transactions.

2.Audit Trail Mechanism

- ✓ **Tracking transaction history** to detect fraud.
- ✓ **Regulatory compliance** for businesses handling crypto.

3.Multisig wallets require multiple signatures to approve a transaction, increasing security.

Backup & Recovery Mechanism

- Wallets generate a **12- or 24-word seed phrase** (BIP39) to restore access if lost.
- Some advanced wallets provide **Shamir's Secret Sharing** for distributed recovery.

Secret key (commonly known as a **private key**) in Bitcoin is a **randomly generated number** that allows users to access and control their Bitcoin holdings. It acts as a **password** to prove ownership of Bitcoin stored in a public address.

📌 What is a Secret (Private) Key?

•A **private key** is a **256-bit random number** used for **signing transactions**.
•It is **kept secret** to prevent unauthorized access to a Bitcoin wallet.
•Every private key has a corresponding **public key** and **Bitcoin address**.

🔹 Example of a Bitcoin Private Key (Hex Format)

5J3mBbAH58C7...X45kmTxVFG1fSND

Goals

availability: You can spend your coins.

security: Nobody else can spend your coins.

convenience

Simplest approach: store key in a file,
on your computer or phone



Very convenient.

As available as your device.

device lost/wiped ⇒ key lost ⇒ coins lost

As secure as your device.

device compromised ⇒ key leaked ⇒ coins stolen

Wallet software

Keeps track of your coins, provides nice user interface.

Nice trick: use a separate address/key for each coin.

- benefits privacy (looks like separate owners)
- wallet can do the bookkeeping, user needn’t know

📌 Centralized Wallets List (2024)

Wallet Name	Company	Custody Type	Supported Cryptos	Security Features	Website
Coinbase	Coinbase, Inc.	Custodial	BTC, ETH, USDT, more	2FA, Cold Storage	coinbase.com
Binance Wallet	Binance	Custodial	BTC, BNB, ETH, more	Multi-Sig, 2FA	binance.com
Kraken Wallet	Kraken	Custodial	BTC, ETH, SOL, more	Cold Storage, API Security	kraken.com
Crypto.com Wallet	Crypto.com	Custodial	BTC, ETH, CRO, more	Multi-Sig, 2FA	crypto.com
Bitstamp Wallet	Bitstamp Ltd.	Custodial	BTC, XRP, ETH, more	Cold Storage, 2FA	bitstamp.net
Huobi Wallet	Huobi Global	Custodial	BTC, ETH, USDT, mo	Hot & Cold Storage	huobi.com

Hot and Cold Storage

1 Cold Secret Keys (Cold Storage)

A **cold secret key** is a **private key that is generated and stored offline**, never touching an internet-connected device.

- **Stored securely** in hardware wallets, paper wallets, air-gapped computers, or secure USB drives.
- **Prevents hacking, phishing, and malware attacks** since it is never exposed online.
- Used for **long-term storage** of large amounts of cryptocurrency.

How It Works:

✅ Generate the private key offline.
✅ Store it securely in a hardware wallet, USB, or paper.
✅ Never expose it to the internet.
✅ Use it only for signing transactions offline.

2 Hot Address for Cold Storage 🔥

A **hot address** is a **public address generated from a cold private key** and used for receiving funds.

- The **public key/address can be shared online**, while the **private key remains offline**.
- The **hot address is used for deposits** into cold storage.
- When needed, **transactions are signed offline** using the cold key and then broadcast online.

How It Works:

✅ Use a cold key to generate a hot address.
✅ Share the hot address to receive payments.
✅ Sign transactions offline using the cold key.
✅ Broadcast the signed transaction online.

2. Hot Secret Keys (Hot Wallets)

A **hot secret key** refers to a private key that is **stored on an online system** and is used to sign transactions automatically.

- **Characteristics of Hot Secret Keys:**
  - Private keys are **stored online**, making them accessible for immediate transactions.
  - Used for frequent transactions, such as exchanges or active wallets.
  - More convenient but **less secure** (vulnerable to hacking).

1. Cold Addresses (Cold Wallets)

A **cold address** refers to a cryptocurrency or asset-holding address that is **not directly connected to the internet**. This is used to store assets securely and protect them from online threats, such as hacking or unauthorized access.

- **Characteristics of Cold Addresses:**
  - Private keys are stored **offline** (e.g., hardware wallets, paper wallets, air-gapped computers).
  - Transactions must be signed **offline** and then broadcasted online.
  - Used for **long-term storage** of assets.
  - More secure but less convenient for frequent transactions.

Hot storage



online

convenient but risky

Cold storage



offline

archival but safer

separate keys

Hot storage



online

Cold storage



offline

hot secret key(s)

cold address(es)

payments



cold secret key(s)

hot address(es)

Hot storage



online

hot secret key(s)

cold address(es)

payments

Cold storage



offline

Problem:

Want to use a new address (and key) for each coin sent to cold

But how can hot wallet learn new addresses if cold wallet is offline?

Awkward solution:

Generate a big batch of addresses/keys, transfer to hot beforehand

Better solution:

Hierarchical wallet

Steps to Generate a Public Key from a Private Key

Bitcoin uses the **secp256k1** elliptic curve to derive the public key.

- ✓ **Generate a random 256-bit private key**
- ✓ **Multiply the private key by the secp256k1 generator point (G)**
- ✓ **Result = Public Key (a coordinate on the curve: (X, Y))**

In Bitcoin, a **public key** is generated from a **private key** using **Elliptic Curve Cryptography (ECC)**. This process is **one-way**, meaning you can derive a **public key** from a **private key**, but **not the other way around**.

```python
import os
import ecdsa

# Step 1: Generate a random private key (32 bytes = 256 bits)
private_key = os.urandom(32)
private_key_hex = private_key.hex()
print(f"🔑 Private Key: {private_key_hex}")

# Step 2: Generate the Public Key (Elliptic Curve Multiplication)
sk = ecdsa.SigningKey.from_string(private_key, curve=ecdsa.SECP256k1)
public_key = sk.verifying_key.to_string()

# Step 3: Convert Public Key to HEX format
public_key_hex = "04" + public_key.hex()  # 04 prefix means uncompressed for
print(f"📢 Public Key: {public_key_hex}")
```

Why Can't We Reverse the Public Key to Get the Private Key?
Bitcoin security relies on the **Elliptic Curve Discrete Logarithm**

Problem (ECDLP), which is **mathematically infeasible** to reverse.
✓ **Forward: Private Key → Public Key (Easy, One Multiplication)**
❌ **Reverse: Public Key → Private Key (Impossible, No Efficient Algorithm)**

Bitcoin uses **public-private key pairs** for security, ownership verification, and transaction signing. Instead of generating keys randomly each time, **Hierarchical Deterministic (HD) wallets** (BIP32/BIP44) provide an efficient way to generate and manage keys securely.

Generate Public-Private Key Pairs?

Reason	Why It's Needed
Ownership & Security	A private key proves ownership of Bitcoin.
Transaction Signing	Bitcoin transactions must be signed using a private key.
Public Key Derivation	Public keys are used to generate Bitcoin addresses.
One-Way Function	Private key → Public key is easy, but reversing it is impossible.

✅ Without private keys, Bitcoin ownership cannot be proven.

🔷 Problems With Traditional Key Generation (Non-HD Wallets)

❌ **Each key is independent** → If you lose a private key, funds are lost forever.

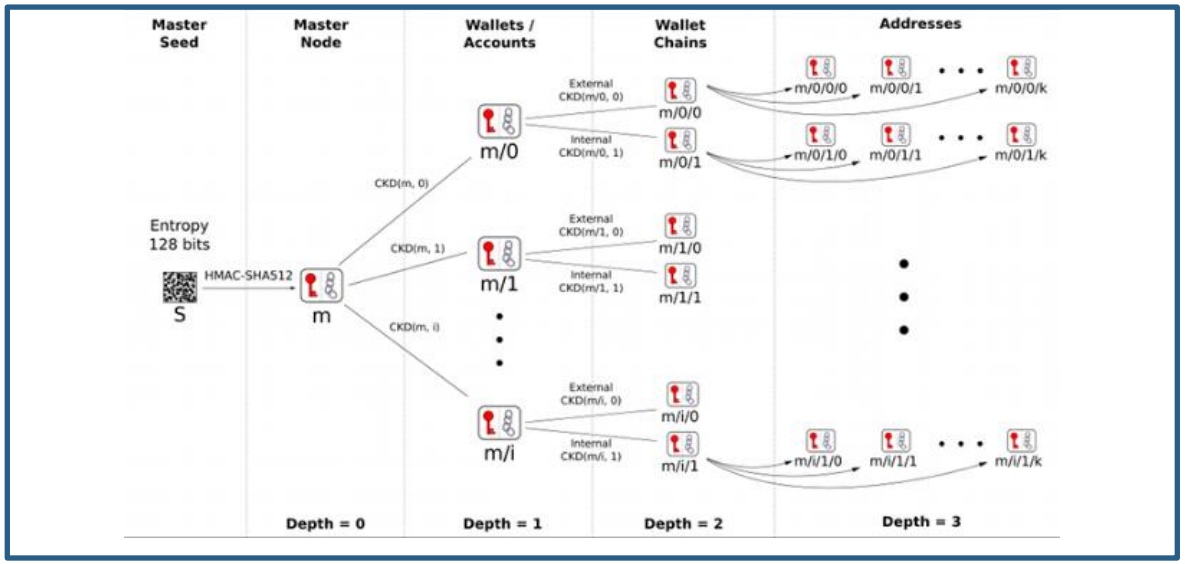
❌ **Hard to manage multiple keys** → Every new address requires a new private key backup.

❌ **No simple backup method** → Need to store each private key separately.

✅ **Solution: HD Wallets!**

🔷 With **HD wallets (BIP32/BIP44)**, all keys come from **one master private key.**

🔷 A **single seed phrase (BIP39 mnemonic)** can recover an entire wallet.



Why Use Hierarchical Key Generation?

✅ **Multiple Child Keys:** You can derive an unlimited number of addresses from one master key.
✅ **Deterministic Key Derivation:** Each child key is mathematically linked to its parent, ensuring consistency.
✅ **Privacy Improvement:** Generating new addresses without manually storing private keys reduces tracking risks.

1. Generating a BIP32 Master Key (Extended Private Key)

If MultiChain supports BIP32, you can generate a master key using `getnewaddress` and `dumpprivkey`:

```bash
multichain-cli chain12 getnewaddress
```

Example Output:

```
1QYBRPLYsQPs4KnNKaG8vL1okmezwtyrW3TzNL
```

Now, retrieve the private key:

```bash
multichain-cli chain12 dumpprivkey 1QYBRPLYsQ...KnNKaG8vL1okmezwtyrW3TzNL
```

2. Generating a BIP32 Public Key (Extended Public Key)

In Bitcoin-like wallets, the **extended public key (xpub)** is used to derive child addresses without exposing private keys. MultiChain does not natively support `getxpub`, but in general BIP32 wallets, you can use:

```bash
bitcoin-cli getxpub "L2o7F6kX5gGAGs9W63g3mdPyd7sPj2qQJbG18NjgHgdC62npy7Pf"
```

This will return an **xpub key**, which can be used for **watch-only wallets**.

3. Deriving Child Keys from a BIP32 Master Key

A BIP32 wallet supports hierarchical derivation like this:

```scss
Master Key (m)
 ├── Child Key 1 (m/0)
 ├── Child Key 2 (m/1)
 ├── Child Key 3 (m/2)
```

You can generate child addresses from the master key using:

```bash
bitcoin-cli deriveaddresses "xpub6CUGRUonZSQ4TWtTMmzXdrXDtypWKiKrB42c.../0/*" 5
```

This will derive 5 addresses from the `xpub`.

📌 3️⃣ Benefits of Using HD Key Generation

Feature	Why It's Important
Single Backup	A 12-word seed can restore all addresses.
Infinite Addresses	One master key can generate unlimited addresses.
Improved Privacy	A new Bitcoin address is generated for each transaction.
Easy Wallet Management	No need to store multiple keys manually.
Better Security	Private keys are never reused.

Key Management in MultiChain

1. **Non-Hierarchical Keys:**
    - MultiChain does not use **BIP32 (HD wallets)**. Each address is independently generated.
2. **Bitcoin-like Address Generation:**
    - It follows a Bitcoin-like approach to generating private/public key pairs but **without hierarchical key derivation**.
3. **Address and Key Storage:**
    - Keys are stored in the wallet file (`wallet.dat`), similar to Bitcoin Core.
4. **Public Key Retrieval:**
    - Use `validateaddress "address"` to fetch the public key.

```
D:\>multichain-cli chain1 validateaddress 1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP
{"method":"validateaddress","params":["1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP"],"id":"83036591-1742200058","chain_name"
"chain1"}

{
    "isvalid" : true,
    "address" : "1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP",
    "ismine" : true,
    "iswatchonly" : false,
    "isscript" : false,
    "pubkey" : "03056b54515296f4cb410a93e401e66538868fd7f6747456c225dd3f83e6507b31",
    "iscompressed" : true,
    "account" : "",
    "synchronized" : true
}
```

```
D:\>multichain-cli chain1 dumpprivkey 1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP
{"method":"dumpprivkey","params":["1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP"],"id":"63705557-1742200353","chain_name":"ch
in1"}

V7oTcT7hXDmfEMdK7ZUMQD2wtJQGNUqgJKgp6q8FGmq4zwYe8tZgme5s

D:\>
```

How HD (hierarchical deterministic**)Key Generation Improves Privacy**
Instead of using the **same Bitcoin address repeatedly**, an HD wallet generates **new addresses dynamically**.

- **Without HD wallets (Single Key Wallets):**

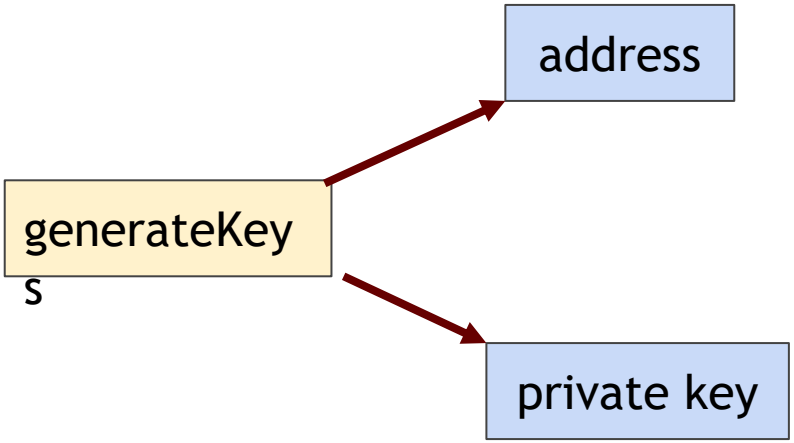
•Every transaction **uses the same Bitcoin address**.
•Anyone can track **all transactions linked to that address**.
•**Privacy risk**: Transactions are publicly visible.

- **With HD wallets (Multiple Derived Addresses):**

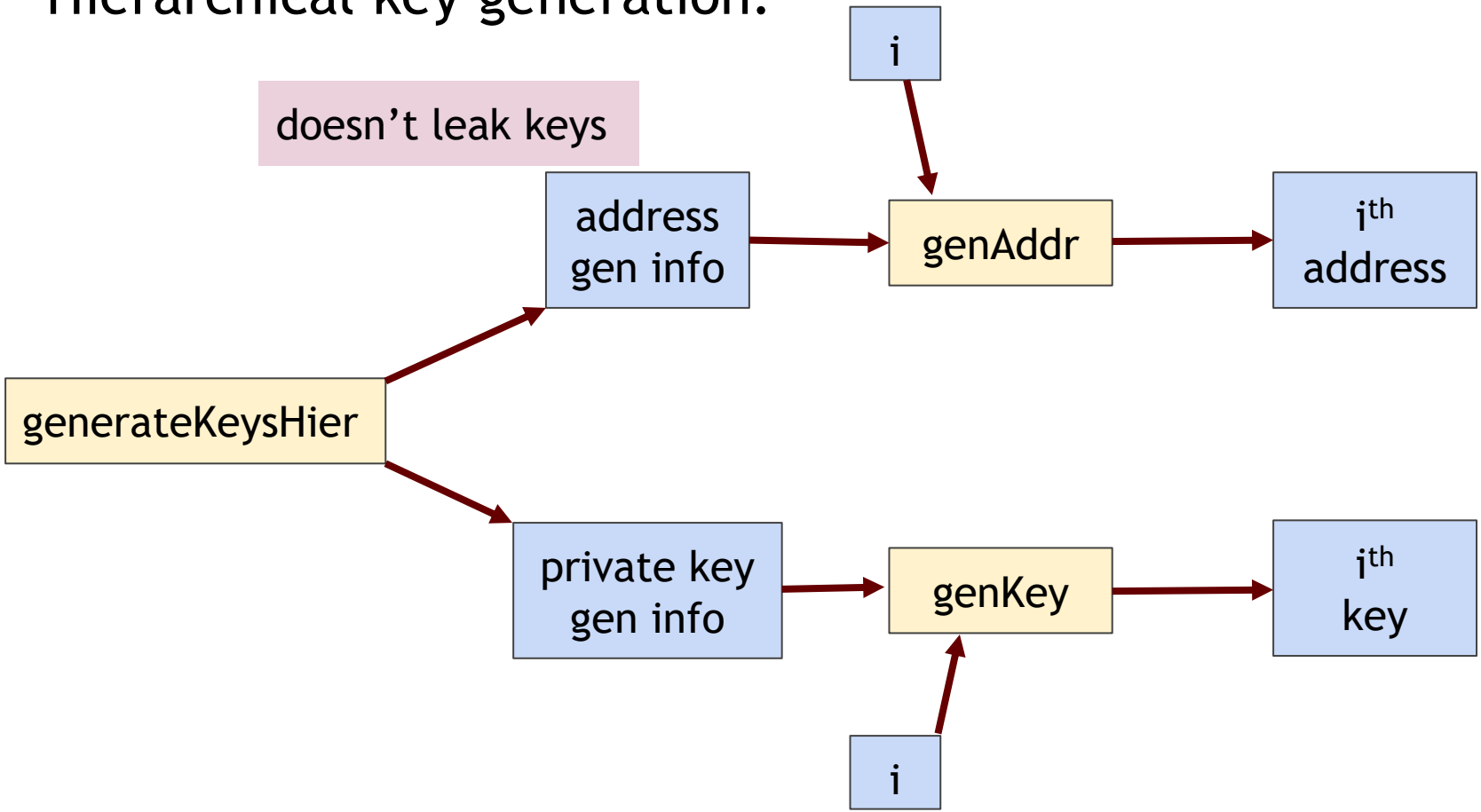
•A new **public key (Bitcoin address) is generated for each transaction**.
•**Balances are split across multiple addresses**.
•**Difficult to track transactions** → Better privacy.

✅ **Bitcoin addresses change automatically, making tracking harder.**

Regular key generation:



Hierarchical key generation:



1️⃣ Hot Side (Hot Wallets) 🔥

- **Definition:** The "hot side" refers to **online or actively used keys**, which are connected to the internet and used for real-time transactions.
- **Examples:**
  - Wallets on **exchanges** (e.g., Binance, Coinbase).
  - Mobile or desktop wallets (e.g., MetaMask, Trust Wallet).
  - **Private keys stored in an internet-connected system.**

Pros ✅

✔ Fast and convenient for transactions.

✔ Suitable for **daily trading and spending.**

Cons ❌

❌ Higher risk of hacking and malware attacks.

❌ If compromised, funds can be stolen instantly.

2 Cold Side (Cold Wallets) ❄️

- **Definition:** The "cold side" refers to **offline key storage**, meaning private keys are never exposed to the internet.
- **Examples:**
  - Hardware wallets (e.g., **Ledger, Trezor**).
  - Paper wallets (printed private keys stored securely).
  - Air-gapped machines (computers never connected to the internet).

Pros ✅

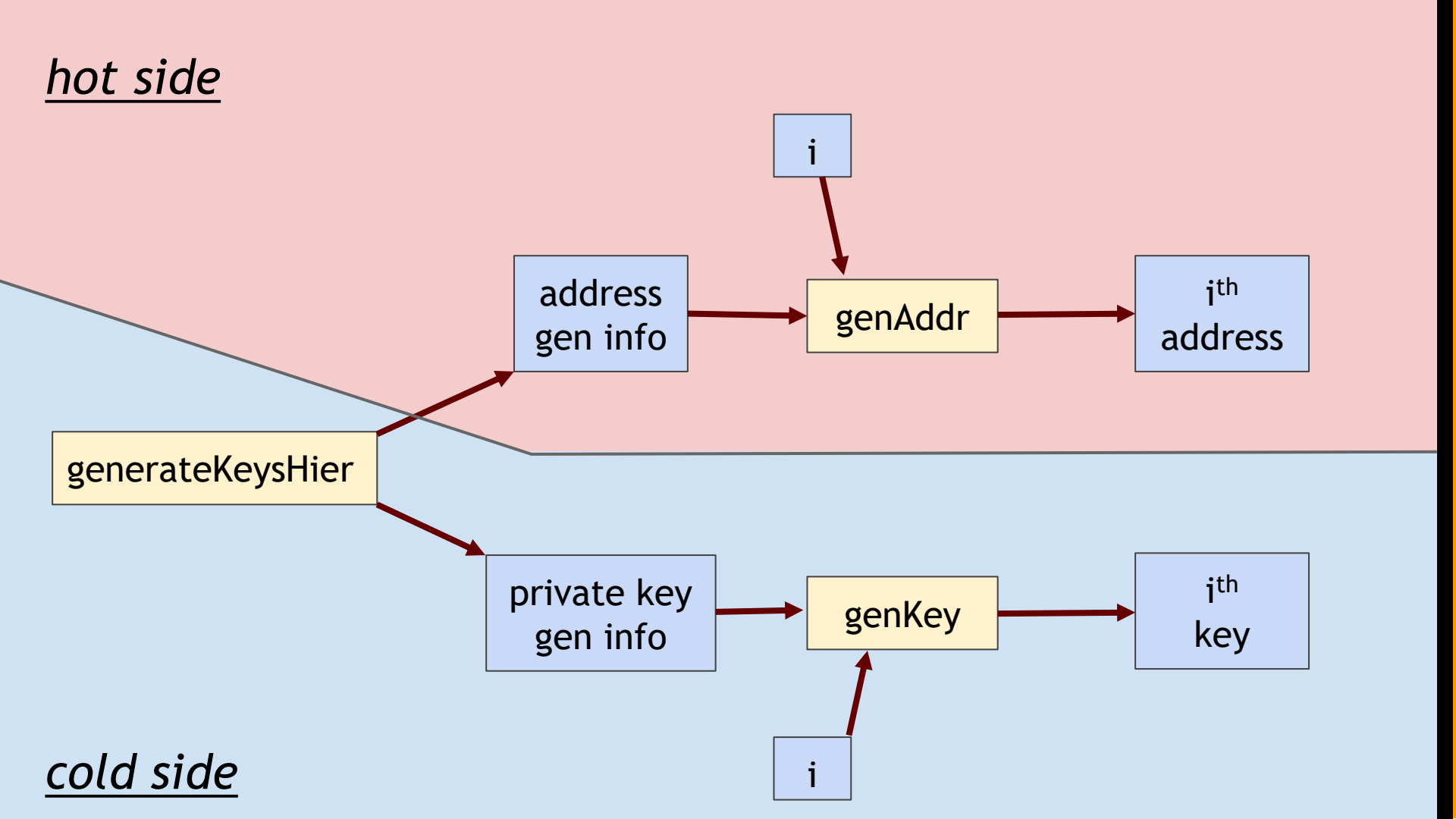
✔ Extremely secure from cyber threats.

✔ Best for **long-term storage of large amounts of crypto.**

Cons ❌

❌ Slower transactions since you need to manually move funds.

❌ Risk of physical damage (if you lose the paper w[illegible]t or hardware device).


hot side
i
address
gen info
genAddr
ith
address
generateKeysHier
private key
gen info
genKey
ith
key
i
cold side

How to store cold info

(1) Info stored in device, device locked in a safe

(2) “Brain wallet”

encrypt info under passphrase that user remembers

(3) Paper wallet

print info on paper,

lock up the paper

(4) In “tamperproof” device

device will sign things for you, but won’t divulge keys

Splitting and Sharing Keys

1 Why Use Secret Sharing for Private Keys?

✅ **Eliminates a Single Point of Failure** – No single party has full control over the private key.

✅ **Improves Security** – Even if one share is compromised, the key remains secure.

✅ **Fault Tolerance** – If some shares are lost, the key can still be reconstructed.

✅ **Multi-Party Control** – Ideal for organizations where multiple approvals are needed for transactions.

2 How Does Secret Sharing Work?

A common technique for secret sharing is **Shamir's Secret Sharing (SSS)**.

Shamir's Secret Sharing (SSS) Algorithm

- The private key is split into **N shares.**
- A minimum of **K out of N** shares are needed to reconstruct the key (**threshold scheme**).
- The shares are distributed to different parties or stored separately.

For example, if you have a **Bitcoin private key** and split it into **5 shares with a threshold of 3 (3/5)**:

- Any **3 out of 5** shares can **recover** the private key.
- If fewer than **3 shares** are available, the private key **remains unrecoverable**.

Secret sharing

Idea: split secret into N pieces, such that

given any K pieces, can reconstruct the secret

given fewer than K pieces, don't learn anything

Example: N=2, K=2

P = a large prime

S = secret in [0, P)

R = random in [0, P)

split:

$X_1 = (S+R) \bmod P$ $X_2 = (S+2R) \bmod P$

reconstruct:

$(2X_1-X_2) \bmod P = S$

Secret sharing

Good: Store shares separately, adversary must compromise several shares to get the key.

Bad: To sign, need to bring shares together, reconstruct the key. ⇐ vulnerable

Multi-sig

Recall multi-sig from Lecture 3.

Lets you keep shares apart, approve transaction without reconstructing key at any point.

Example

Andrew, Arvind, Ed, and Joseph are co-workers.
Their company has lots of Bitcoins.

Each of the four generates a key-pair,
puts secret key in a safe, private, offline place.

The company's cold-stored coins use multi-sig, so that three of the four keys must sign to release a coin.

Online Wallets and Exchanges

Online wallet

like a local wallet

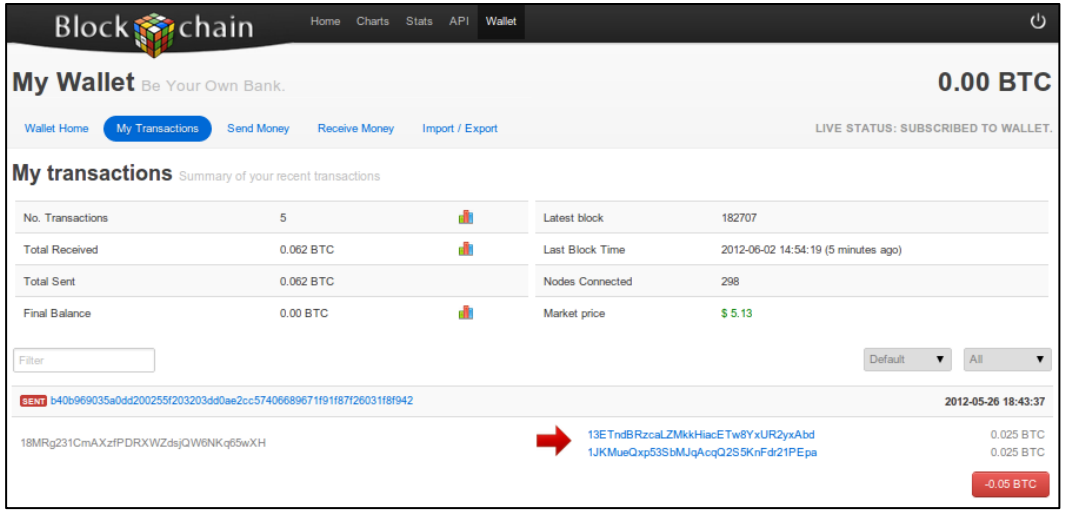
but “in the cloud”

runs in your browser

site sends code

site stores keys

you log in to access wallet



Online wallet tradeoffs

convenient: nothing to install, works on multiple devices

but security worries

vulnerable if site is malicious or compromised

ideally, site is run by security professionals

Bank-like services

you give the bank money (a “deposit”)

bank promises to pay you back later, on demand

bank doesn’t actually keep your money in the back room

- typically, bank invests the money
- keeps some around to meet withdrawals (“fractional reserve”)

Bitcoin Exchanges

accept deposits of Bitcoins and fiat currency ($, €, …)

promise to pay back on demand

lets customers:

make and receive Bitcoin payments

buy/sell Bitcoins for fiat currency

typically, match up BTC buyer with BTC seller

What happens when you buy BTC

suppose my account at Exchange holds $5000 + 3 BTC
I use Exchange to buy 2 BTC for $580 each

result: my account holds $3840 + 5 BTC

note: no BTC transaction appears on the blockchain
only effect: Exchange is making a different promise now

Exchanges: Pros and Cons

pro: connects BTC economy to fiat currency economy
easy to transfer value back and forth

con: risk
same kinds of risks as banks



6 issues for £9 + FREE iPad & iPhone editions

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Study: 45 percent of Bitcoin exchanges end up closing

TECHNOLOGY / 26 APRIL 13 / by IAN STEADMAN

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A study of the Bitcoin exchange industry has found that 45 percent of exchanges fail, taking their users' money with them. Those that survive are the ones that handle the most traffic -- but they are also the exchanges that suffer the greatest number of cyber attacks.



Almost half of all exchanges close *Shutterstock*

Computer scientists Tyler Moore (from the Southern Methodist University, Dallas) and Nicolas Christin (of Carnegie Mellon University) found 40 exchanges on the web which offered a service of changing bitcoins into other fiat currencies or back again. Of those 40, 18 have gone out of business -- 13 closing without warning, and five closing after suffering security breaches that forced them to close. Four other exchanges have

Bank Regulation

for traditional banks, government typically:

- imposes minimum reserve requirements
  - must hold some fraction of deposits in reserve
- regulates behavior, investments
- insures depositors against losses
- acts as lender of last resort

Proof of Reserve

Bitcoin exchange can prove it has fractional reserve.

fraction can be 100%

Prove how much reserve you're holding:

publish valid payment-to-self of that amount

sign a challenge string with the same private key

Prove how many demand deposits you hold: ...

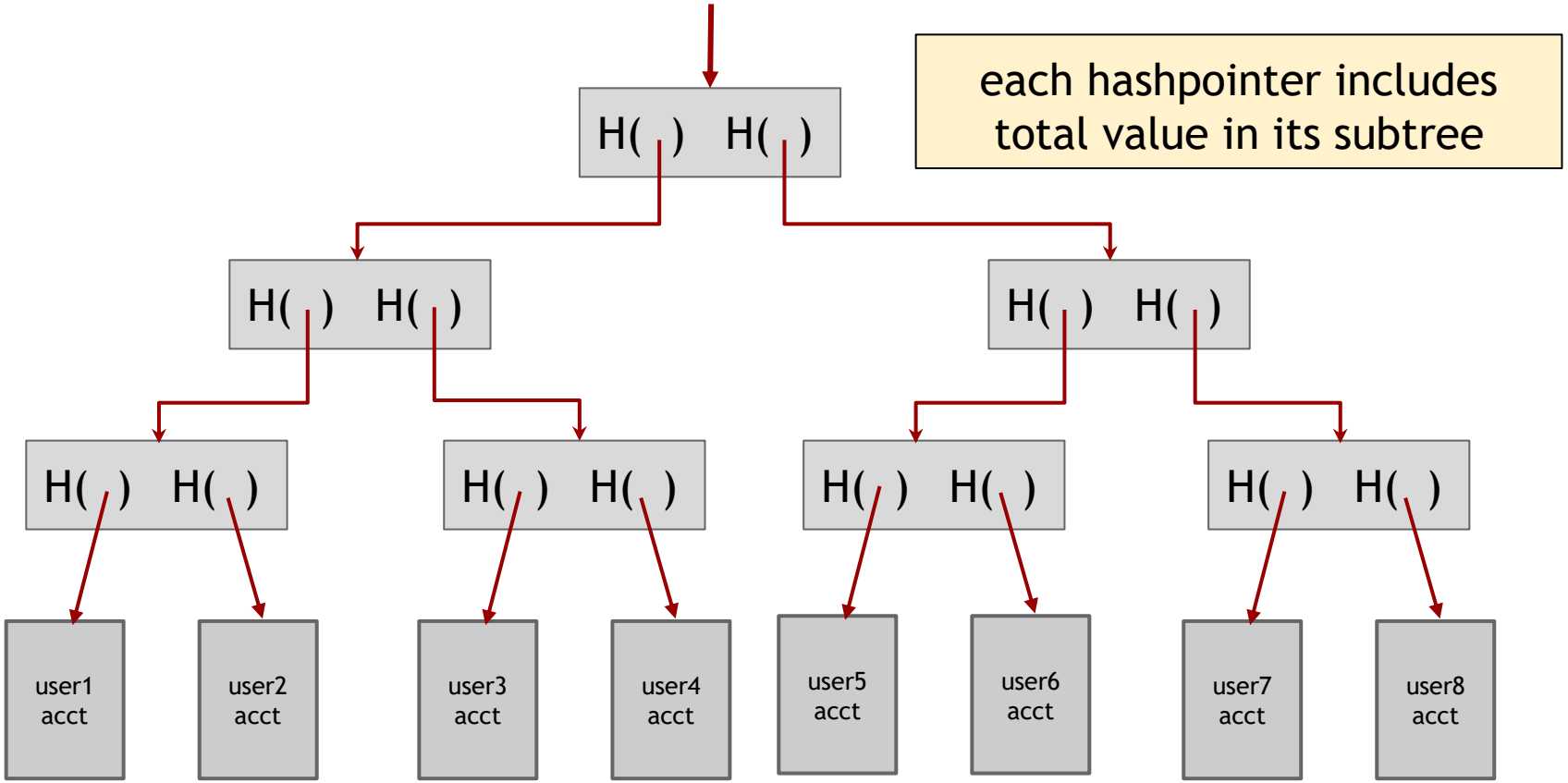
```
D:\>multichain-cli chain1 issue "1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP" "USDToken" 1000000 0.01
{"method":"issue","params":["1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP","USDToken",1000000,0.01],"id":"31827753-1742205686
,"chain_name":"chain1"}

f55e403c16f75b6ead99d85fb2fde47ff51b597aac4a12e5b740385f5c5d3236

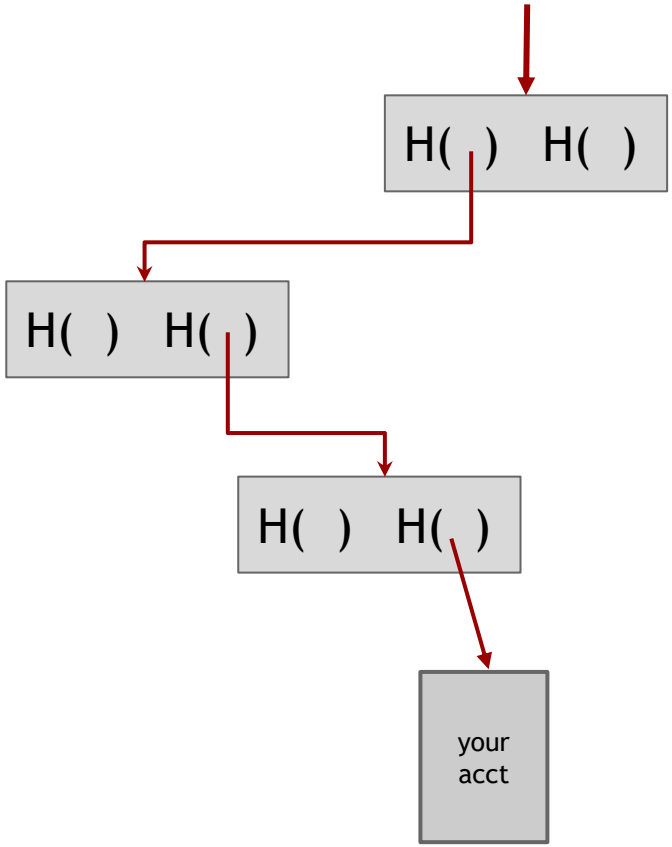
D:\>multichain-cli chain1 getaddressbalances 1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP
{"method":"getaddressbalances","params":["1J7GRPaUciy4abqDRKrWWsJ7WLJQHR7qwggcLP"],"id":"73340861-1742205695","chain_na
e":"chain1"}

[
    {
        "name" : "USDToken",
        "assetref" : null,
        "qty" : 1000000
    }
]
```

Merkle tree with subtree totals



Checking that you're represented in the tree



show O(log n) items

Proof of Reserve

Prove that you have at least X amount of reserve currency

Prove that customers have at most Y amount deposited

So reserve fraction ≥ X / Y

```
C:\>multichain-cli chain12 getaddressbalances 1QYBRPLYsQPs4KnNKaG8vL1okmezwtyrW3TzNL
{"method":"getaddressbalances","params":["1QYBRPLYsQPs4KnNKaG8vL1okmezwtyrW3TzNL"],"id":"46574602-1742213208","chain_name":"chain12"}

[
    {
        "name" : "PakCoin",
        "assetref" : "60-265-58129",
        "qty" : 1000000
    },
    {
        "name" : "sana",
        "assetref" : "151-266-43807",
        "qty" : 1000
    },
    {
        "name" : "asset3",
        "assetref" : "71-266-50882",
        "qty" : 70
    },
    {
        "name" : "asset4",
        "assetref" : "72-265-8226",
        "qty" : 80
    },
    {
        "name" : "rashid",
        "assetref" : "138-267-48398",
        "qty" : 60
    },
```

```
C:\>multichain-cli chain12 listassets
{"method":"listassets","params":[],"id":"29644154-1742213471","chain_name":"chain12"}

[
    {
        "name" : "PakCoin",
        "issuetxid" : "11e3d06a8d0ee4858fd18f7d1a4ffc86b9e489d92f4c67b0b438785f10dc7fa7",
        "assetref" : "60-265-58129",
        "multiple" : 100,
        "units" : 0.01,
        "open" : false,
        "restrict" : {
            "send" : false,
            "receive" : false,
            "issue" : true
        },
        "fungible" : true,
        "canopen" : false,
        "canclose" : false,
        "totallimit" : null,
        "issuelimit" : null,
        "details" : {
        },
        "issuecount" : 1,
        "issueqty" : 1000000,
        "issueraw" : 100000000,
        "subscribed" : false
```

K-Electric implements a **Multichain blockchain** to manage and verify customer **service charge deposits, payments, and reserves** using **Proof of Reserve (PoR)**.

1. **Customer Prepayment & Asset Creation**
    - ✓ Customers deposit service charges in advance (e.g., **500 KEL-Tokens**)
    - ✓ K-Electric issues an equivalent asset (KEL-Tokens) on the blockchain, proving that the balance is reserved.

Command in Multichain(Sample Command)

multichain-cli chain1 issuefrom "KE_wallet" "KEL-Tokens" 500 1 0

2. **Customer Consumption & Deduction**
    - ✓ As customers consume electricity, K-Electric deducts **KEL-Tokens** accordingly.
    - ✓ If a customer uses electricity worth **200 KEL-Tokens**, the updated balance is verified.

Transfer example:

multichain-cli chain1 sendasset "customer_wallet" "KEL-Tokens" 200

1. **Verifying Proof of Reserve**

   - ✓ At any point, customers can check how much **KEL-Tokens** remain in their account.
   - ✓ K-Electric publicly displays its reserve to ensure it can cover all customer balances.

Check total reserves in K-Electric's wallet:

multichain-cli chain1 gettotalbalances

2. **Auditing & Transparency**

   - ✓ Regulatory authorities and consumers can audit K-Electric's reserves to ensure there's no overcharging.
   - ✓ A smart contract can **automate refunds** if K-Electric overcharges a customer.

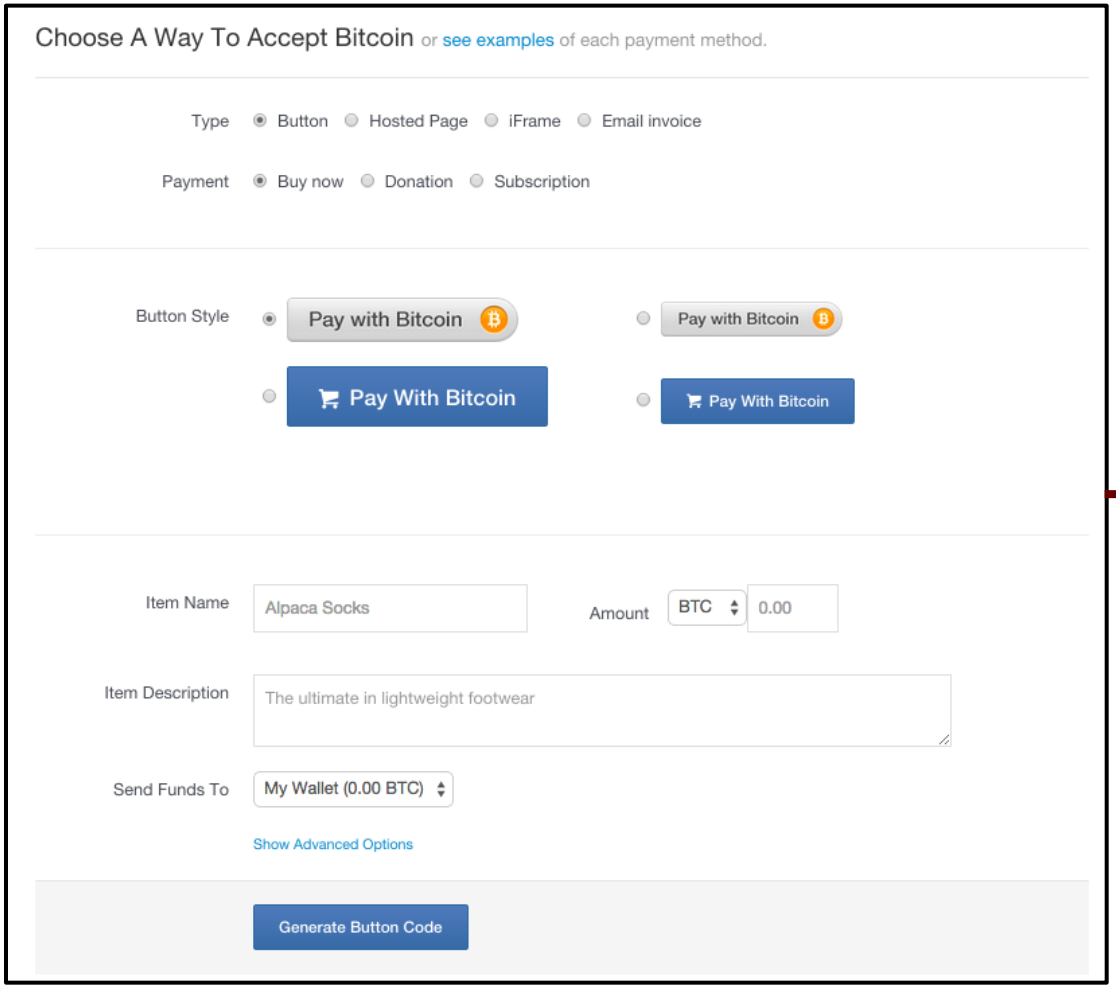
Payment Services

Scenario: merchant accepts BTC

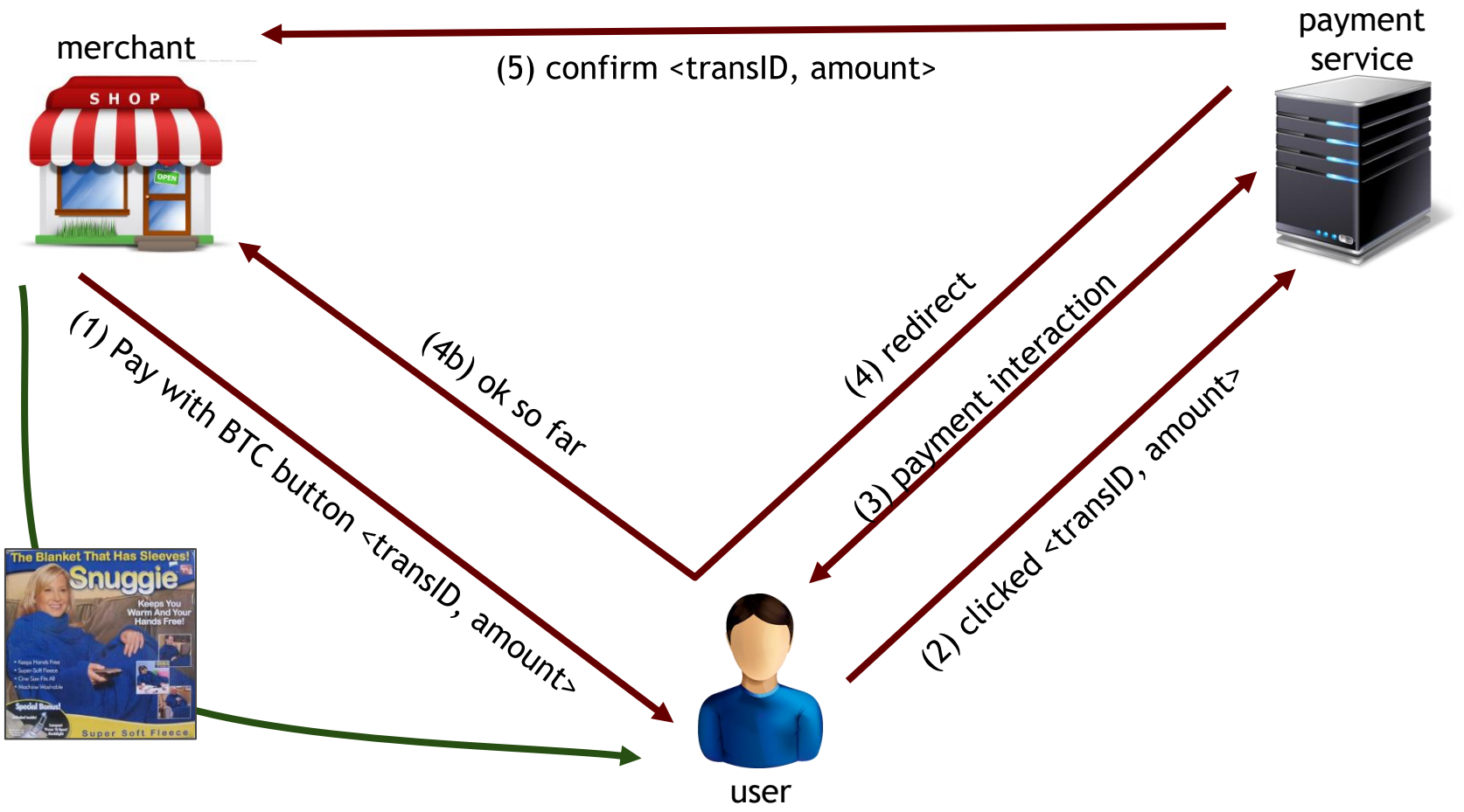
customer wants: to pay with Bitcoin

merchant wants:

* to receive dollars
* simple deployment
* low risk (tech risk, security risk, exchange rate risk)



HTML for payment button


merchant
payment service
(5) confirm <transID, amount>
(4b) ok so far
(4) redirect
(3) payment interaction
(2) clicked <transID, amount>
(1) Pay with BTC button <transID, amount>
user

End result

customer: pays Bitcoins

merchant: gets dollars, minus a small percentage

payment service:

- gets Bitcoins
- pays dollars (keeps small percentage)
- absorbs risk: security, exchange rate
- needs to exchange Bitcoins for dollars, in volume

Transaction Fees

Recall:

transaction fee = value of inputs - value of outputs

fee goes to miner who records the transaction

Interesting economics, discussed in later lecture

How are transaction fees set today?

Costs resources for

peers to relay your transaction

miner to record your transaction

Transaction fee compensates for (some of) these costs

Generally, higher fee means transaction will be forwarded and recorded faster.

Current consensus fees:

No fee if

tx less than 1000 bytes in size,

all outputs are 0.01 BTC or larger, and

priority is large enough

Priority = (sum of inputAge*inputValue) / (trans size)

Otherwise fee is 0.0001 BTC per 1000 bytes

Approx transaction size: $148\ N_{inputs} + 34\ N_{outputs} + 10$

Most miners enforce the consensus fee structure.

If you don’t pay the consensus fee, your transaction will take longer to be recorded.

Miners prioritize transactions based on fees and the priority formula.

Currency Exchange Markets

http://bitcoincharts.com/markets

Symbol	Latest Price	30 days	Average	Volume	Low/High	Bid	Ask	24h Avg.	Volume	Low/High
▼ BitStamp USD bitstampUSD	582.54 2 min ago		620.52 -37.98 -6.12%	155,811.67 96,683,593.16 USD	570.5 658.88	581.13	582.54	585.63 -3.09 -0.53%	6,189.14 3,824,569.60 USD	574.15 596
Bitfinex USD bitfinexUSD	619.78 6 days, 5 hrs ago		632.10 -12.32 -1.95%	126,042.21 79,671,138.43 USD	593.37 665	579.31	580.49	—	0.00 0.00 USD	— —
▼ btc-e USD btceUSD	572.78 0 min ago		615.51 -42.73 -6.94%	106,578.66 65,599,931.43 USD	562 654.381	572.541	572.779	576.33 -3.55 -0.62%	3,396.32 1,957,406.31 USD	566.001 585.85
▼ itBit USD itbitUSD	581.69 just now		618.36 -36.67 -5.93%	34,726.55 21,473,457.56 USD	571 662	580.27	581.11	582.64 -0.95 -0.16%	1,607.07 936,342.67 USD	577 587.99
▲ ANX USD anxhkUSD	593.43896 29 min ago		624.73 -31.29 -5.01%	30,902.66 19,305,871.63 USD	565.166 687.21424	577.2	593.34886	587.47 5.97 1.02%	1,476.78 867,565.29 USD	565.3373 602.06006
▲ LocalBitcoins USD localbtcUSD	977.52 9 min ago		665.75 311.77 46.83%	17,221.75 11,465,390.41 USD	492.94 2529.6	1163.78	558.61	636.33 341.19 53.62%	840.60 534,896.62 USD	531.87 2500
1coin USD 1coinUSD	605.3 4 days, 6 hrs ago		625.85 -20.55 -3.28%	14,973.92 9,371,488.64 USD	601.5 664.5	605.1	605.3	—	0.00 0.00 USD	— —
▼ hitbtc USD hitbtcUSD	583.41 0 min ago		622.80 -39.39 -6.32%	14,778.51 9,203,987.87 USD	573.23 657.47	581.54	583.33	587.71 -4.30 -0.73%	459.21 269,883.25 USD	576.72 594.67
▲ CoinTrader USD cotrUSD	589.76 31 min ago		619.79 -30.03 -4.85%	1,460.39 905,136.91 USD	0.1 700	580	588.37	585.16 4.60 0.79%	76.52 44,773.46 USD	580.66 599.68
▼ Camp BX USD cbxUSD	593 1 hr, 36 min ago		633.51 -40.51 -6.40%	1,062.60 673,170.82 USD	585.14 670	595	604	606.28 -13.28 -2.19%	36.03 21,844.39 USD	585.14 626.8
▼ Ripple USD rippleUSD	583.71244672 6 min ago		621.33 -37.62 -6.05%	567.36 352,513.96 USD	574.98 655.99	582.03	585.71244671	584.69 -0.97 -0.17%	18.40 10,757.11 USD	575.6908721 590.9794998
▲ Kraken USD krakenUSD	586.5 18 min ago		625.75 -39.25 -6.27%	169.82 106,263.87 USD	574.57864 658.87046	586.5	597.75871	583.67 2.83 0.49%	1.37 800.09 USD	574.57864 591.90124
▼ bitKonan USD bitkonanUSD	581 2 hrs, 48 min ago		624.21 -43.21 -6.92%	99.32 61,997.57 USD	551 668	581.08	615	605.10 -24.10 -3.98%	2.43 1,467.97 USD	581 610
▲ The Rock Trading Company USD rockUSD	581 0 min ago		613.09 -32.09 -5.23%	77.86 47,734.81 USD	575 699.99	587.24	604.91	578.77 2.23 0.39%	2.15 1,244.36 USD	575 581
▼ Justcoin USD justUSD	579.92 16 hrs ago		624.54 -44.62 -7.14%	59.56 37,197.01 USD	578.113 700	614.93	631.21	589.41 -9.49 -1.61%	0.30 175.70 USD	578.197 599.999
▲ BitBay USD bitbayUSD	586.57 4 hrs, 57 min ago		609.30 -22.73 -3.73%	58.04 35,361.52 USD	547 631.12	586.44	588.17	586.45 0.12 0.02%	1.17 688.35 USD	583.98 586.57
▲ Vircurex USD vcxUSD	620.00124 8 hrs, 57 min ago		620.23 -0.23 -0.04%	3.76 2,329.85 USD	590 710	621	648	601.61 18.39 3.06%	0.05 30.40 USD	590 620.00124

Basic market dynamics

market matches buyer and seller

large, liquid market reaches a consensus price

price set by supply (of BTC) and demand (for BTC)

Supply of Bitcoins

supply = coins in circulation (+ demand deposits?)

coins in circulation: fixed number, currently ~13.1 million

When to include demand deposits?

When they can actually be sold in the market.

Demand for Bitcoins

BTC demanded to mediate fiat-currency transactions

- Alice buys BTC for $
- Alice sends BTC to Bob
- Bob sells BTC for $

BTC "out of circulation" during this time

BTC demanded as an investment

- if the market thinks demand will go up in future

Simple model of transaction-demand

T = total transaction value mediated via BTC ($ / sec)

D = duration that BTC is needed by a transaction (sec)

S = supply of BTC (not including BTC held as long-term investments)

$\frac{S}{D}$ Bitcoins become available per second

$\frac{T}{P}$ Bitcoins needed per second

Equilibrium:

$$P = \frac{TD}{S}$$